Final Examination

Econ 103, Statistics for Economists

May 4th, 2017

You have 120 minutes to complete this exam. Graphing calculators, notes, and textbooks are not permitted.

I pledge that, in taking and preparing for this exam, I have abided by the University of Pennsylvania's Code of Academic Integrity. I am aware that any violations of the code will result in a failing grade for this course.

Name: _____

Student ID #: _____

Signature: _____

Question:	1	2	3	4	5	6	Total
Points:	35	35	25	40	45	60	240
Score:							

Instructions: Answer all questions in the space provided, continuing on the back of the page if you run out of space. Show your work for full credit but be aware that writing down irrelevant information will not gain you points. Be sure to sign the academic integrity statement above and to write your name and student ID number on *each page* in the space provided. Make sure that you have all pages of the exam before starting.

Warning: If you continue writing after we call time, even if this is only to fill in your name, twenty-five points will be deducted from your final score. In addition, two points will be deducted for each page on which you do not write your name and student ID.

35 1. Mars inc. produces the "colorful button-shaped chocolates" M&M's. The contents of a bag of M&M's changed in 1995 when tan M&M's were replaced by blue M&M's. The relative frequencies of the remaining colors changed as well. The values you will need to solve this problem appear in bold in the following table:

	blue	tan	**green**	orange	**yellow**	**red**	brown
Before 1995 (*Old*)	–	10%	**10%**	10%	**20%**	**20%**	30%
After 1995 (*New*)	24%	–	**20**%	16%	**14%**	**13%**	13%

I have two bags of M&M's: one from before 1995 (*Old*) and one from after 1995 (*New*). I randomly choose a bag, such that each is equally likely to be selected. I then make three independent random draws with replacement from the bag. I obtain: green, yellow, red. Given this information, what is the probability that I selected the *Old* bag?

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2. The $\chi^2(m)$ is a random variable that we did not study in lecture this semester. If $Z_1, \ldots, Z_m \sim$ iid $N(0,1)$ then $Y = Z_1^2 + Z_2^2 + \cdots + Z_m^2$ is a $\chi^2(m)$ RV. In other words the $\chi^2(m)$ RV is the *sum of squares* of m iid standard normal RVs. The $\chi^2(m)$ RV has a single parameter: the *degrees of freedom* m. R has a built-in function `rchisq` for making random draws from a $\chi^2(m)$ distribution. In this question we will create our own version: `myrchisq`. In your answers you may use any R functions *except* `rchisq`.

[10] (a) Create an R function called `draw_chisq` that constructs a single draw from a $\chi^2(m)$ random variable by making m iid standard normal draws and calculating their sum of squares. Your function should take a single input argument – the degrees of freedom m – and return the $\chi^2(m)$ random draw.

[10] (b) Create an R function called `myrchisq` that repeatedly calls `draw_chisq(m)` to generate n iid draws from a $\chi^2(m)$ distribution. It should take two inputs – the number `n` of χ^2 draws, and degrees of freedom `m` – and return a vector of n iid $\chi^2(m)$ draws.

[10] (c) Write R code that uses `myrchisq(n,m)` to approximate the probability that a $\chi^2(1)$ RV takes on a value strictly greater than 4 using 10,000 Monte Carlo simulations.

[5] (d) Using what you know about the standard normal RV, approximately what would be the numeric result of running your code from part (c)?

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3. Let $Y_1, Y_2, Y_3 \sim$ iid $N(0, \sigma^2 = 36)$ and define: $Z = Y_1 + \frac{Y_2}{2} + \frac{Y_3}{3}$.

[5] (a) Calculate $E[Z]$

[5] (b) Calculate $\text{Var}(Z)$

[5] (c) What kind of random variable is Z? Specify the values of any and all parameters.

[10] (d) For what value of c is $P(Z < c) \approx 0.975$? Your answer should specify a numeric value and not rely on any R commands.

The following question is taken verbatim from your Homework for Lectures 16–17.

4. This question is based on a recent paper examining how "organic" labeling changes people's perceptions of different food products. Researchers recruited volunteers at a local mall in Ithaca, New York and gave each two samples of yogurt to taste. Although both yogurts were in fact identical, the volunteers were *told* that one of them was organic while the other was not. After tasting both, each volunteer was asked to estimate how many calories each of the samples of yogurt contained. (Since, unknown to the volunteer, both samples contained exactly the same kind of yogurt, each in fact contained the same number of calories.) To prevent confounding from anchoring or other behavioral effects, the order in which a given volunteer tasted the two yogurts, i.e. "organic" first or "organic" second, was chosen at random. The results of this experiment are stored in an R data table called `yogurt`. Here are the first few rows:

```
> head(yogurt)
  regular organic
1      60       40
2       5        0
3     200      100
4      60       40
5     100      100
6      90       90
```

Each row in this data table corresponds to a single individual's guess of the number of calories contained in each of the two yogurts. For example, the values 60 and 40 in row 1 mean that volunteer number one guessed that the regular yogurt sample contained 60 calories and the organic sample contained 40. Summary statistics for the two columns are as follows:

	`regular`	`organic`
Sample Mean	113	90
Sample Var	3600	2916
Sample SD	60	54
Sample Corr.	0.8	
Sample Size	115	

8 (a) Give the units of each of the summary statistics from above:

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Sample Mean _____
Sample Var. _____
Sample SD _____
Sample Corr. _____

6 (b) Sara thinks that this experiment should be analyzed as independent samples data. Assume that she is correct and construct an approximate 95% CI for the difference of means (`regular` - `organic`) based on the CLT.

6 (c) Kevin thinks that this experiment should be analyzed as matched pairs data. Assume that he is correct and construct an approximate 95% CI for the difference of means (`regular` - `organic`) based on the CLT.

6 (d) How do the confidence intervals constructed by Sara and Kevin differ? What is the reason for this difference? Who has has constructed the appropriate confidence interval for this example: Kevin or Sara? Explain briefly.

6 (e) Suppose that Kevin wanted to carry out a two-sided test of the null hypothesis that organic labeling does not affect consumer's estimates of caloric content, on average. What is his test statistic? What R command should he use to calculate the p-value for his test? Will his result be greater or less than 0.05?

8 (f) Using your knowledge of experiments, observational studies, hypothesis testing, and confidence intervals, what conclusions can we draw from this study? Explain briefly.

5. Let $X_1, \ldots, X_n \sim$ iid Bernoulli(p) and define $\widehat{p} = \sum_{i=1} X_i/n$. This question concerns a test of $H_0\colon p = 0.5$ against $H_1\colon p > 0.5$ with $\alpha = 0.025$.

[5] (a) If $p = 0.5$ and n is large, what is the approximate sampling distribution of $\widehat{p}$?

[5] (b) Write down the test statistic T_n for the test specified in the problem statement. Be sure to fully impose the null hypothesis.

5 (c) If $p = 0.5$ and n is large, approximately what is the sampling distribution of the test statistic from part (b)? Using this information, what is the critical value and decision rule for the test in the problem statement?

5 (d) If $p = 0.5$ what is the probability that we will reject H_0? Explain briefly.

5 (e) If $p = 0.8$ and n is large, what is the approximate sampling distribution of $\widehat{p}$?

10 (f) Based on your answer to part (e), if $p = 0.8$ and n is large, what is the approximate sampling distribution of the test statistic T_n constructed in part (b)?

10 (g) Continuing from part (f), if $p = 0.8$ for what value of n is the power of the test in the problem statement approximately equal to 0.84?

6. The data table `companyX` contains a random sample of 215 employees from Company X:

```
   Male Months Salary
1:    1     91  69250
2:    1     22  53120
3:    0     40  57280
4:    1     88  69830
5:    1     21  56470
6:    0     46  54890
```

Each row is an employee: `Male` takes on the value 1 if a given employee is male and zero otherwise, `Months` gives total months of work experience, and `Salary` gives annual

salary in dollars. To answer this question you will need to consult the regression results and plots on the final two pages of this exam. You may want to tear these pages out for convenience. For full credit, be sure to clearly reference the specific set of regression results you rely on in each of your answers below. These are numbered 1–4.

[4] (a) Write R code to add a new column to `companyX` called `Years` that gives work experience in *years* rather than months: e.g. 22 months becomes 1.83 years.

[4] (b) Write the R code needed to generate the boxplots of `Salary` and `Years` found on the final page of this exam. You do not have to label the plots.

[3] (c) Based on the regression results how many years of work experience do the female employees at Company X have on average?

[3] (d) Based on the regression results how much do the male employees at Company X earn per year on average?

[4] (e) Suppose you had to choose between using `Male` or `Years` to predict the salary of an employee at Company X. Which appears to give more accurate predictions for the employees in our sample? How much more accurate? Explain briefly.

[6] (f) Suppose we want to test the null hypothesis that male and female employees earn the same salary, on average, against the two-sided alternative with $\alpha = 0.05$. Do

we reject or fail to reject? Explain briefly and show all of your work for full credit.

4 (g) Write R code to make a scatterplot with `Salary` on the y-axis and `Years` on the x-axis and plot the corresponding regression line on top of the points.

3 (h) Approximately what is the correlation between `Salary` and `Experience`?

3 (i) What are the units of the slope estimate in Regression #3?

3 (j) What are the units of the *standard error* of the slope estimate in Regression #3?

3 (k) Is the intercept in Regression #3 a meaningful quantity? If not, why not? If so, what does it mean? Explain briefly.

20 (l) Use the statistical tools you have learned in Econ 103 to explain the evidence of a pay gap between male and female employees using the results of Regression #4. Write your answer in bullet points with *no more than five* bullets. Clear and succinct

responses will be graded more favorably than long, rambling ones.

Regression #1

```
lm(formula = Salary ~ Male, data = companyX)
            coef.est coef.se
(Intercept) 62059.27   981.92
Male        -3159.65  1405.07
---
n = 215, k = 2
residual sd = 10298.43, R-Squared = 0.02
```

Regression #2

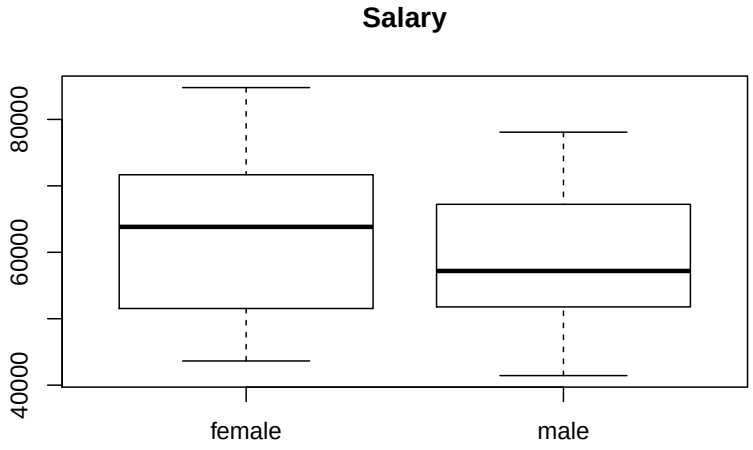
```
lm(formula = Years ~ Male, data = companyX)
            coef.est coef.se
(Intercept)  5.44     0.27
Male        -1.17     0.39
---
n = 215, k = 2
residual sd = 2.86, R-Squared = 0.04
```

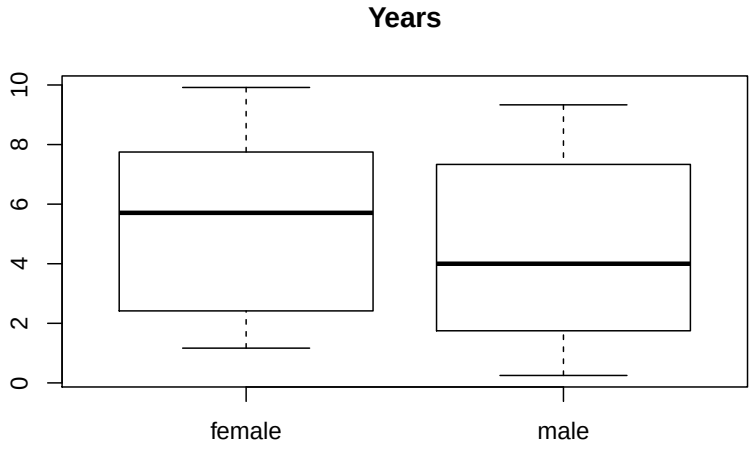
Regression #3

```
lm(formula = Salary ~ Years, data = companyX)
            coef.est coef.se
(Intercept) 43940.75   418.48
Years        3406.18    73.85
---
n = 215, k = 2
residual sd = 3143.62, R-Squared = 0.91
```

Regression #4

```
lm(formula = Salary ~ Male + Years + Male:Years, data = companyX)
              coef.est coef.se
(Intercept)   41377.92   613.04
Male           4413.82   799.88
Years          3804.79   100.23
Male:Years     -734.92   141.38
---
n = 215, k = 4
residual sd = 2947.67, R-Squared = 0.92
```





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